

## Elia Jean Botros

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### EDUCATION

**The Pennsylvania State University, University Park**

Expected Graduation: December 2026

*Bachelor of Science in Industrial Engineering*

**GPA: 3.3**

*Minor: Engineering Leadership Development*

### SKILLS & CERTIFICATIONS

Software and General Skills: MATLAB, Python, SQL, Excel, SolidWorks, AutoCAD, Simio, Prototwin, Microsoft Project, Minitab, LINDO, Figma, Process Flow Mapping, Time & Motion Studies

Certification: Lean Six Sigma Green Belt - IISE (Expected End of Sept. 2026)

### WORK EXPERIENCE

**Kew Gardens Animal Hospital, Queens, New York**

**May 2026 - Aug. 2026**

*Animal Hospital Assistant*

- Applied process standardization to surgical prep workflows, reducing average instrument setup time and improving procedure throughput across a high-volume clinical environment.
- Designed and maintained standardized equipment and inventory protocols.
- Optimized appointment scheduling and patient intake processes to balance daily caseload, reduce bottlenecks, and minimize patient wait times.

### ENGINEERING PROJECTS

**Cardinal Systems Automation Feasibility Study, Schuylkill Haven, PA**

**Aug. 2026 - Present**

*Project Lead & Process Analyst*

- Coordinating directly with the sponsor's President, CFO, and Director of Operations to define project scope and deliverable milestones.
- Conducted on-site time studies across manual work stations, breaking the process into work elements to establish a cycle time baseline and identify bottleneck operations.
- Evaluating integration feasibility of four idle Yaskawa Motoman GP180 robots against reach, payload, and layout constraints while modeling cost and payback versus the current manual line.

**ACRP University Design Competition**

**Jan. 2025 - May 2025**

*Team Lead & System Designer*

- Led system design of GSE Sync, an AI-powered fleet routing system using reinforcement learning and shortest-path algorithms to optimize 250+ ground support vehicles at Seattle-Tacoma International Airport, targeting a 60% reduction in vehicle idle time.
- Collaborated with a 4 member team to develop MATLAB and SUMO simulation prototypes using real OpenStreetMap airport data to model dynamic vehicle routing across multiple flight delay scenarios.
- Performed risk analysis and developed mitigation strategies across identified risk categories.

### LEADERSHIP & INVOLVEMENT

**Mentor & Tutor, FIRST Robotics Competition Team 2638**

**Jan. 2024 - May 2024**

- Provided remote technical mentorship to FIRST Robotics Team 2638, advising on mechanical systems design, CAD modeling in SolidWorks, and systems-level integration.
- Guided students through engineering design tradeoffs, manufacturing feasibility, and competition requirements through virtual collaboration.

**Broadcast Lead, Penn State Intramural Sports**

**Jan. 2023 - Oct. 2023**

- Led live broadcast operations coordinating technical setup, production flow, and real-time problem solving during events.